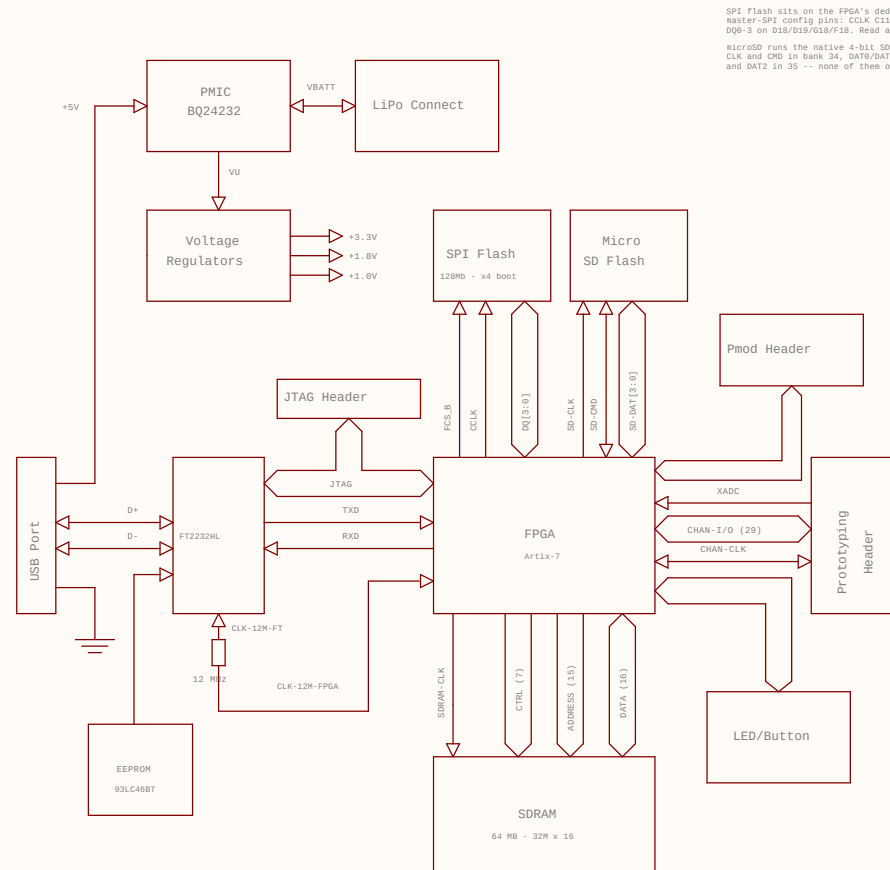
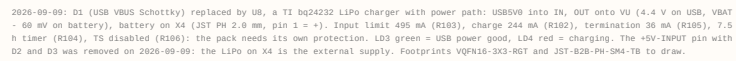
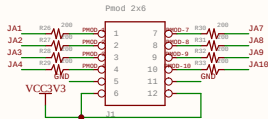
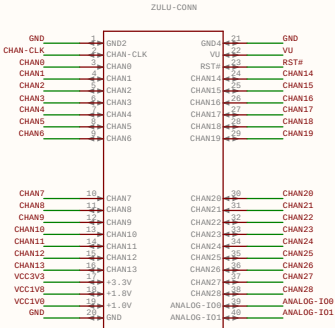




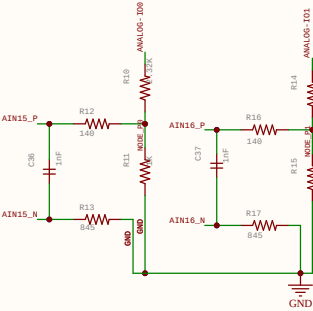
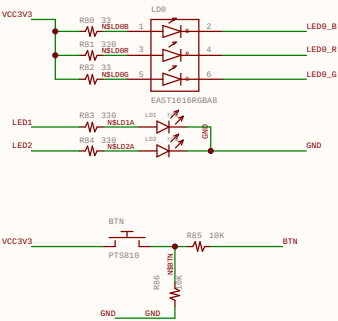


Power Supplies			
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X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images. TOP ROW, x 50.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), then CHAN7-CHAN13 (10-16), +3.3V, +1.8V, +1.0V, GND (17-20), then CHAN14-CHAN16 (21-23), CHAN17-CHAN19 (24-26), CHAN20-CHAN28 (29-36), ANALOG-IO0, ANALOG-IO1 (37, 38), CHAN17-CHAN19 (27-29), then CHAN20-CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40). FOUR positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 9 and 10 above and between pins 29 and 30 below -- are RESERVED, not spare: the edge-mounted micro-USB X1 lands on the top row (7.88 mm of copper), the top-side LiPo header X4 (JST B2B-PH-SM4-TB, 7.95 mm body) lands on the bottom row directly opposite it.

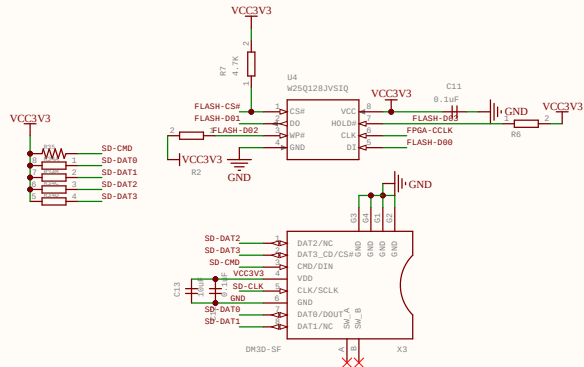


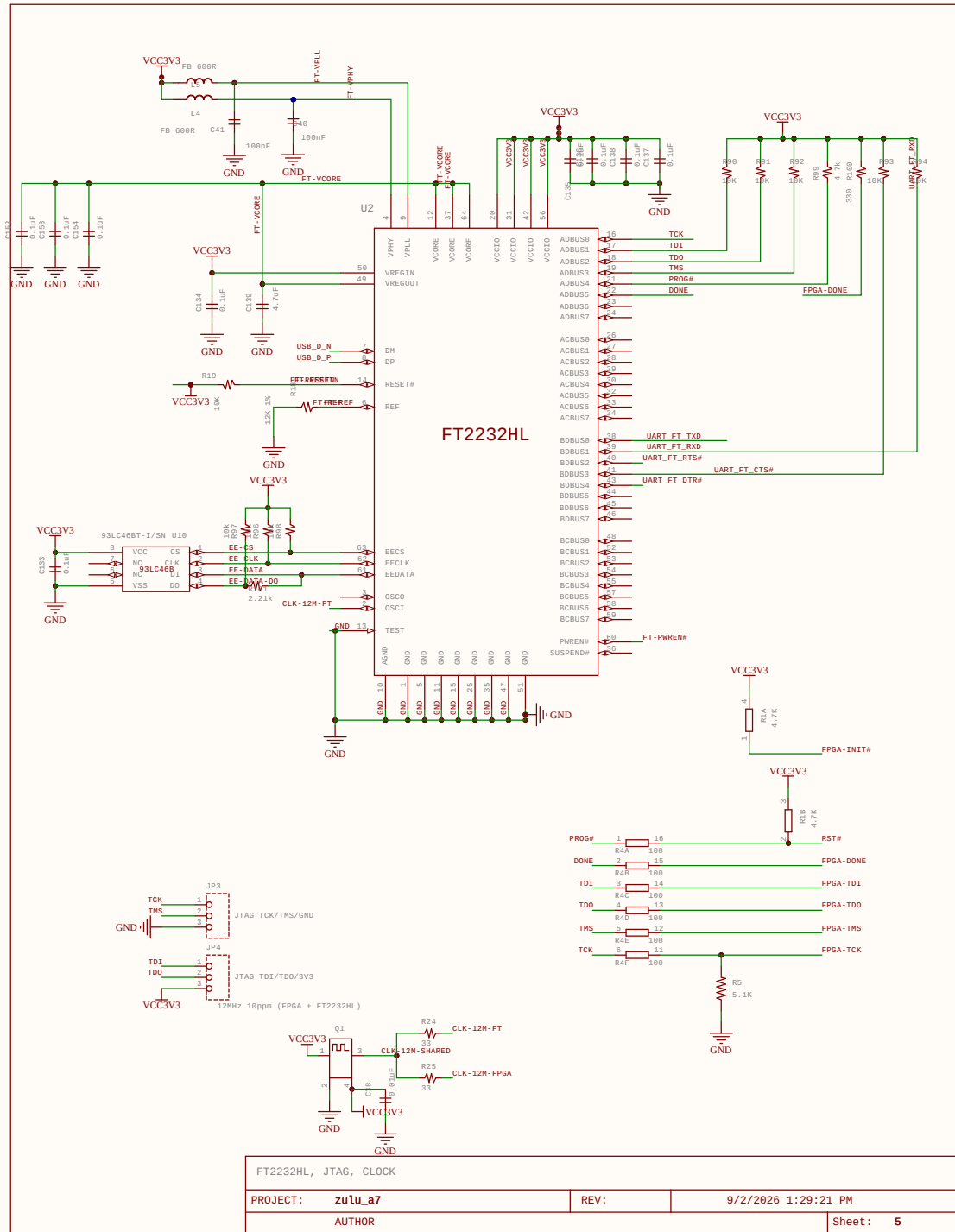
Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmd A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16, A17 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



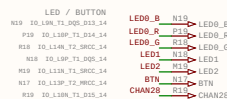
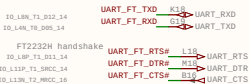
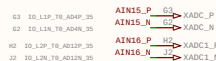
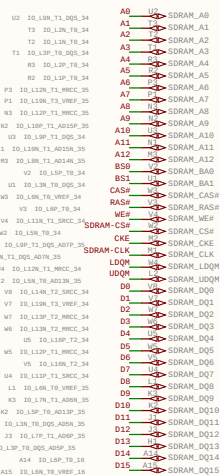
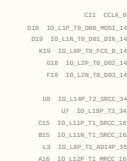
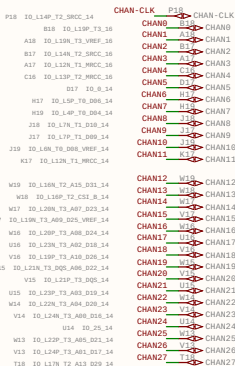
General IO			
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MASTER SPI BOOT (x4). R2/R1:R0 - 0:0:1 -- R2B pulls WB to VCC3V3, R21/R22 pull M1/M2 to GND -- so at power-up the FPGA drives CLK itself and clocks its bitstream out of U4 with no USB and no host present. That mode can only use the dedicated configuration pins: CLK C11, FCS_B K19, D00_M0S1 D18, D02_D1N D19, D02 G18, D03 F18. No other ball will do, which is why SDRAM A6 had to move off X19 (it is now on D17, still bank 14). After DONE the six pins revert to ordinary user I/O. R2A/R2B keep WP#/HOLD# high through the x1 command phase and through the float window before the FPGA drives D02/D03 (PUDC_B is tied high, so the FPGA has no internal pull-ups). R2D keeps FCS_B deselected. Vivado: set_property CONFIG_MODE SPIx4 [current_design] set_property BITSTREAM.CONFIG SPI_BUSWIDTH 4 [current_design] The microSD is deliberately NOT on this bus any more: an inserted card sits in native SD mode watching bitstream traffic go past on what it believes is CMD, and one 48-bit window with a valid CRC7 is enough to make it answer and drive the line the FPGA is reading. X3 now has its own CLK/CMD/DAT0 on R2/T1/T2, the bank-34 balls the flash just vacated, with R9 damping DAT0 where the socket stub is.





PROOFREADING HEADER



FPGA Connections			
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